Nikhil Kalidasu

University of Texas at Austin (GPA: 3.6)
BS, Computer Science; BS, Biology

Portfolio: **nik875.github.io** **Email:** **srikarnikhil.kalidasu@gmail.com** **Phone: +1 571-435-9126**

Coursework: Data Structures, Algorithms & Complexity, Software Engineering, Operating Systems, Genetics, Data Science, Cell Biology, Randomized Algorithms, Computer Vision, Statistics & Probability, Cloud Computing.

Skills: Python, R, C/C++, Java, JavaScript, PyTorch, TensorFlow, CNN, GNN, Transformers, LLMs, Linux, Bash, NumPy, SQL, Postgres, Pandas, REST APIs, Docker, AWS, GCP, Reinforcement Learning (RL), Bayesian Statistics, HPC.

Work Experience

Collegiate Cyber Defense Competition – *Infrastructure & Security* *(Jan 2026 – Apr 2026)*

- Managed **15+ machines**, maintained SQL/Postgres databases, REST APIs, Docker containers, AWS infrastructure.
- Won **first place** for service uptime and red team suppression and **second place** overall.

Sessions Lab, Cal Tech – *Undergraduate Research Assistant* *(Jun 2025 – Jan 2026)*

- NRIQ: **Bayesian optimization** for predicting kinetic isotope effects in microbial sulfate reduction, TCA cycle.
- Implemented **parallelization** for 28-core HPC, evaluated NRIQ against baselines, developed **user-friendly UI**.

AeroParagon (Counter-Drone Startup) – *AI/ML Engineering Intern (Computer Vision)* *(Jan 2025 – Mar 2025)*

- **Fine-tuned RetinaNet CNN**, optimizing with **Neural Architecture Search (NAS)** for edge deployment.
- **Productionized auto-labeling pipeline** for 3 GB videos via motion detection with **MoG** and **Kalman filter**.

De Anda Lab, University of Florida – *Computational Scientist* *(Aug 2024 – Aug 2025)*

- Developed **SeREGen**, machine learning embedding generator for DNA built on **Transformer, CNN, PyTorch**.
  - Used to visualize **40 novel bacterial phyla** and **COVID-19 spike protein** mutations for vaccine resistance.

ECLAIR Robotics – *Project Lead* *(Aug 2023 – Apr 2025)*

- **Led a ~25 person team** to develop an autonomous RC car from scratch, self-driving algorithm runs on **Raspberry Pi**.
  - Evolved autonomous driving agents with **Genetic Algorithm + PPO reinforcement learning** to handle sparse rewards.
- Developed JavaScript music recommendation app with fine-tuned Huggingface LLM **(~500M params)**.

Baker Lab, UT Austin – *Undergraduate Research Assistant* *(Apr 2024 – Aug 2024)*

- Authored **$100K** research grant, earned **4800 SU** compute grant for HPC distributed supercomputer.
- Developed CLI plugin with ChatGPT that **enhanced productivity,** reduced learning curve via natural-language interface.

Emerging Diagnostic and Investigative Technologies (EDIT) Lab – *ML Intern* *(Jun 2021 – Aug 2021)*

- Developed Convolutional **(CNN)** tumor classifier, Graph Neural Network **(GNN)** to remove ink markings from H&E images.
- Performed **spatial domain analysis** on 16GB multimodal H&E + RNAseq data with Graph Convolution **(GCN).**

Publications

Adversarial License Plate Rim *(Aug 2025 – Jun 2026)*

- Adversarial border degrades license plate reader accuracy by 60%. **First author publication:** https://arxiv.org/abs/2604.02457, **accepted** to Security in Machine Learning Applications workshop at ACNS conference 2026.
- **Self-initiated**: sourced collaborators, managed cloud infrastructure across three providers, and wrote manuscript independently.

Blockchain Strategies for Real-World Governance *(May 2025 – Aug 2025)*

- Can proof-of-stake voting improve democratic governance? **First author preprint:** https://ssrn.com/abstract=5393983